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Lab 03 - NAND Only Logic

In this lab, you've learned how to convert arbitrary logical equations into NAND only circuits, and why that might be a good thing.

Rubric

Item	Description	Value
Summary Answers	Your writings about what you learned in this lab.	25%
Question 1	Your answers to the question	25%
Question 2	Your answers to the question	25%
Question 3	Your answers to the question	25%

Lab Summary

Today we learned that you can create an logic circuit as a NAND only system. We translated the formula into a diagram, then replicated on a physical breadboard. We ran into the issues with the switches not functioning consistently, generating false outputs. We resolved those issues by adjusting our logic.

Lab Questions

1 - Write down DeMorgan's Law and the truth tables proving it out.

$$(A+B)' = A' \cdot B'$$

A	B	$F((A+B)')$	A	B	$F((A' \cdot B)')$
0	0	1	0	0	1
0	1	0	0	1	0
1	0	0	1	0	0
1	1	0	1	1	0

$$(A \cdot B)' = A' + B'$$

A	B	$F((A \cdot B)')$	A	B	$F(A' + B')$
0	0	1	0	0	1

0	1	1	0	1	1
1	0	1	1	0	1
1	1	0	1	1	0

2 - What is the value in converting circuits to NAND only?

They are more efficient than other logic circuits and use less components. Consolidating to one gate-type is also very useful.

3 - How does what you did in lab with the breadboard relate to the FPGA?

The breadboard functions as an FPGA where the switches can control the output.

Code Submission

Upload a .zip of all your code or a public repository on GitHub.